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Indirect Costs of Financial Distress in Durable Goods Industries: The Case of Auto Manufacturers

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1 Big picture

- **Question.** Does a manufacturer's financial distress reduce demand for its durable goods, even before bankruptcy actually occurs?
- **Core mechanism.** Durable goods are bundled with future complementary services: warranties, replacement parts, maintenance, dealer networks, repairs, and software or service support. If the producer becomes financially distressed, consumers worry these future services may disappear or become less valuable.
- **Setting.** The paper studies used cars sold in U.S. wholesale auctions from January 1, 2006 to November 14, 2008. The data contain more than 6 million completed transactions with detailed vehicle, auction, and location characteristics.
- **Distress measure.** Manufacturer financial distress is measured by five-year credit default swap spreads. CDS spreads proxy for the market's assessment of default probability.
- **Identification idea.** The paper compares prices of the same detailed car type within the same region and week as the manufacturer's CDS spread moves. The main specification uses car-type-region-week fixed effects.
- **Main result.** A 1,000-basis-point increase in a manufacturer's CDS spread lowers the average used car price by about \$68, or roughly 0.5% of the average transaction price.
- **Warranty mechanism.** The effect is much stronger for cars still under factory warranty and for cars with more of the original warranty remaining. A car with a full factory warranty loses about \$139 per 1,000-basis-point CDS increase.
- **Service-life mechanism.** The price effect is concentrated among low-mileage cars, especially those below roughly 50,000 miles. High-mileage cars, which rely less on future manufacturer services, are much less exposed.
- **Robustness.** The result is not driven by unrelated placebo CDS spreads, dealer financial distress, new car fire sales, new car prices, or several supply-side controls.
- **Economic magnitude.** A 3,000-basis-point increase in GM's CDS relative to Ford implies a large indirect cost: approximately a 6% decrease in GM car margins or a 10% drop in the value of the General Motors North America vehicles division.
- **Capital-structure implication.** The estimated indirect distress costs can be large enough to offset debt tax benefits. The simple trade-off calculation suggests that GM and Ford frequently had leverage above the static optimum.
- **Economic takeaway.** Financial distress is costly not only because of legal or restructuring costs. In durable goods industries, distress can destroy product-market value through consumers' fear that future complementary services will fail.

2 Data and measurement

2.1 Why used car auctions are the right laboratory

- Used car prices move quickly because wholesale auto auctions are liquid and frequent.
- Auction participants are professional dealers, so prices should incorporate information about future retail demand.
- A fall in used car auction prices does not directly reduce a manufacturer's cash flow in the same way that cutting new car prices would. This helps reduce reverse-causality concerns.
- The auction environment provides rich controls: car type, model year, trim, mileage, condition, auction location, auction date, sale format, and repeated wheel-ins.

Interpretation. Used cars are a clean place to observe how consumers and dealers revalue the manufacturer's brand when financial distress threatens future service and warranty support.

2.2 Auction transaction data

- The data come from a large multinational auctioneer with 83 U.S. auction sites.
- The sample covers January 1, 2006 through November 14, 2008.
- The final matched data contain 6,188,759 successful transactions.
- The total value of sales is about \$89 billion, with an average transaction price around \$13,000.
- Each vehicle is identified by VIN, which gives make, model, body style, trim, and model year.
- The auctioneer also records odometer mileage, condition rating, auction format, run number, number of times the car was wheeled through, and whether the buyer was in-lane, online, or upstream.

Interpretation. The richness of the data allows the authors to compare very similar vehicles sold close together in time, rather than relying on aggregate brand-level price movements.

2.3 Vehicle characteristics

- Ford and Chevrolet dominate the sample, representing 22.43% and 15.87% of observations.
- The most common models include Taurus, Explorer 4WD V6, Impala, Altima, Grand Prix, Focus, and F150 pickups.
- The most common model years are 2006 and 2005.
- The most common categories are SUVs and midsize cars.
- Condition 3 vehicles are the largest group, accounting for 58.54% of observations. Condition 4 accounts for 21.17%.

Interpretation. The sample is broad, but it is heavily weighted toward common mass-market vehicles. This is useful because the indirect-cost channel should matter most for ordinary consumers and dealers who rely on manufacturer warranties and parts availability.

2.4 Condition and mileage

- Condition ratings run from 0 to 5.
- Condition 0 vehicles are essentially salvage or parts-only vehicles.
- Condition 5 vehicles have no or only minor defects.

- Average prices rise with condition: condition 0 vehicles average about \$3,743, while condition 5 vehicles average about \$19,086.
- Mileage is divided into twenty non-overlapping mileage bands with equal-sized empirical bins.
- Low-mileage cars are much more valuable: band 1 averages 4,819 miles and \$19,823, while band 20 averages 152,057 miles and \$3,243.

Interpretation. Condition and mileage proxy for the remaining service life of the vehicle. They are also central to the mechanism tests because warranties and future manufacturer support matter more for cars expected to remain in service longer.

2.5 Manufacturer CDS spreads

- The paper uses daily five-year CDS spreads from Thomson Financial DataStream.
- CDS spreads are matched to vehicle transactions using manufacturer identity.
- The average matched manufacturer CDS spread is 643 basis points, with a standard deviation of 856 basis points.
- GM and Ford CDS spreads rise sharply during the 2008 financial crisis.
- Toyota and Honda also experience increases, but from much lower levels.

Interpretation. CDS spreads are used because they are more liquid and timely than corporate bond prices. They capture market beliefs about default risk and therefore the expected risk that warranties and service obligations may be disrupted.

2.6 Warranty measures

The paper uses two warranty variables:

- **In-warranty indicator:** equals one if the car remains within both the age and mileage limits of its original factory warranty.
- **Fraction of remaining warranty:** the minimum of the remaining mileage fraction and the remaining age fraction, with a value of zero once either limit is exceeded.

Interpretation. If the indirect-cost channel works through fear that warranties will not be honored in bankruptcy, then the CDS effect should be stronger for cars with more warranty protection remaining.

2.7 Auction-side controls

The main controls include:

- mileage-bin indicators;
- car condition indicators;
- auction format indicators;
- number of times the car was wheeled through the auction lane;
- run number;
- number of same-trim cars sold that day;
- number of same-trim cars already sold earlier that day.

Interpretation. These controls absorb supply-side congestion, auction sequencing, local inventory pressure, and physical vehicle quality.

3 Models and empirical specifications

3.1 Mechanism in one line

Manufacturer distress $\Rightarrow$ higher probability of warranty/parts/service disruption $\Rightarrow$ lower expected durable-good service flow
(1)

Interpretation. The mechanism is demand-side and forward-looking. Buyers reduce willingness to pay today because the future flow of complementary services becomes riskier.

3.2 Baseline price specification

The core regression is

$$p_{ijklt} = \beta \text{CDS}_{it} + X'_{ijklt} \theta + a_{ijkT} + \varepsilon_{ijklt}. \quad (2)$$

- i indexes manufacturer.
- j indexes detailed car type: model, trim, and model year.
- k indexes auction location or region.
- l indexes the specific auction.
- t indexes day.
- T indexes week.
- a_{ijkT} is a car-type-region-week fixed effect.
- CDS_{it} is the manufacturer's CDS spread in basis points.

Interpretation. The coefficient β is identified from high-frequency within-week variation in CDS spreads and auction prices for the same detailed vehicle type in the same region.

3.3 Normalized price specification

The paper also uses a normalized price outcome:

$$\tilde{p}_{ijklt} = \frac{p_{ijklt}}{\bar{p}_j}, \quad (3)$$

where $\bar{p}_j$ is the average sample price for car type j .

Interpretation. Normalizing asks whether the effect is large relative to the typical value of the vehicle type. This prevents high-price cars from dominating the economic interpretation.

3.4 Service-life interaction specification

For warranty, mileage, or condition measure Z_{ijklt} , the canonical interaction model is

$$p_{ijklt} = \beta \text{CDS}_{it} + \gamma Z_{ijklt} + \delta (Z_{ijklt} \times \text{CDS}_{it}) + X'_{ijklt} \theta + a_{ijkT} + \varepsilon_{ijklt}. \quad (4)$$

Interpretation. If Z measures expected future service flows, the durable-goods hypothesis predicts $\delta < 0$: financial distress should reduce prices more for cars with more future manufacturer-dependent service value.

3.5 Warranty valuation

For cars with a full warranty remaining, Table 8 implies approximately

$$\frac{\partial p}{\partial \text{CDS}} \approx -0.139, \quad (5)$$

so a 1,000-basis-point CDS increase reduces price by about

$$1000 \times 0.139 = \$139. \quad (6)$$

Extrapolating from no bankruptcy risk to certain bankruptcy with a 10,000-basis-point CDS change gives a rough full-warranty valuation of

$$10,000 \times 0.139 = \$1,390. \quad (7)$$

Interpretation. This magnitude is of the same order as manufacturers' accounting accruals for warranty costs, which makes the estimates economically plausible.

3.6 Mileage-channel specification

Mileage is captured by twenty quantile indicators. The paper estimates

$$p_{ijklt} = \beta \text{CDS}_{it} + \sum_{b=2}^{20} \delta_b (\text{CDS}_{it} \times 1\{\text{Band}_{ijklt} = b\}) + X'_{ijklt} \theta + a_{ijkT} + \varepsilon_{ijklt}. \quad (8)$$

Interpretation. The total CDS effect in mileage band b is $\beta + \delta_b$. The expected pattern is most negative for low-mileage cars that still depend on warranties, parts, and service networks.

3.7 Robustness specifications

The paper runs several robustness tests:

- placebo CDS regressions using unrelated manufacturers' CDS spreads;
- tests of whether GM and Ford dealers change used-car purchases when their affiliated manufacturer becomes distressed;
- controls for contemporaneous new car prices;
- controls for expected future new car prices;
- controls for recalls and supply-side movements.

Interpretation. These tests target alternative explanations: reverse causality, fire sales of new cars, distress of affiliated dealers, and confounding demand or supply shocks.

3.8 Indirect cost of financial distress

The paper converts price effects into a firm-value effect. The core marginal relation is

$$\frac{dE(CFD)}{d\text{CDS}} \approx \frac{3.5\%}{1000 \text{ bp}}, \quad (9)$$

where CFD is the cost of financial distress.

For capital-structure calculations, the marginal cost of leverage is written as

$$MC = \frac{dE(CFD)}{d\text{CDS}} \times \frac{d\text{CDS}}{dD}, \quad (10)$$

where D is debt or leverage.

Interpretation. The first term is estimated from the used-car price effect. The second term maps leverage or credit-rating changes into CDS changes.

4 Identification and empirical design

4.1 What identifies the main effect

- The main identifying variation is within detailed car-type-region-week cells.
- The comparison is between similar cars sold in the same region and week while manufacturer CDS spreads change.
- The design absorbs low-frequency brand reputation, general macro shocks, regional demand, vehicle model quality, and week-specific conditions.
- Standard errors are clustered by manufacturer-month to allow serial and cross-vehicle correlation in CDS-related shocks.

Interpretation. The empirical design is demanding: the main coefficient is not driven by Ford versus Toyota in levels, but by short-run movements in distress and prices after very granular fixed effects.

4.2 Why used cars help with reverse causality

- A fall in new car demand can directly hurt manufacturer cash flows and raise distress.
- A fall in used car auction prices does not directly lower the manufacturer's revenue from new car sales.
- Used cars still reflect expected consumer demand for the brand's durable service flow.

Interpretation. The used-car market helps separate the effect of financial distress on product value from the effect of product demand on financial distress.

4.3 Why warranty heterogeneity matters

- If the coefficient simply reflected broad brand reputation, all cars from the distressed manufacturer should respond similarly.
- If the mechanism is future services, cars with more warranty and longer expected life should respond more.
- The paper finds the stronger effect exactly where the service-flow channel predicts it: in-warranty and low-mileage cars.

Interpretation. The warranty and mileage heterogeneity is the most persuasive mechanism evidence in the paper.

4.4 Supply-side concerns

- Rental fleets, manufacturers, or dealers might change the supply of used cars when a manufacturer becomes distressed.
- The paper controls for local auction inventory of same-trim cars and how many have already been sold that day.
- It also controls for auction format, wheel-ins, and run number.
- Robustness tests address new-car fire sales and affiliated dealer distress.

Interpretation. The authors try to ensure that the CDS coefficient is a demand-side valuation effect rather than an artifact of changing auction supply.

4.5 What the paper cannot fully prove

- Manufacturer distress is not randomly assigned.
- CDS spreads can reflect many pieces of information, including future demand, not just bankruptcy probability.
- Warranty disruption is inferred from heterogeneity rather than observed directly for each buyer.
- The main data are wholesale auction prices, so retail consumer preferences are observed through dealer bidding.
- The results are strongest for the auto industry; generalization to all durable goods depends on how important manufacturer-provided complementary services are.

Interpretation. The paper's strength is the combination of very granular transaction data, high-frequency CDS variation, and mechanism-specific heterogeneity tests.

5 Tables and figures

The figures and tables below are directly cropped from the original paper and grouped to save space.

5.1 Tables 1–2: vehicle characteristics and condition codes

Read:

- Ford and Chevrolet are the two largest makes in the sample.
- SUVs and midsize cars are the largest categories.
- Condition 3 vehicles account for more than half of the sample.
- The condition scale runs from salvage/parts-only to near-pristine vehicles.

Economic message: The sample contains a broad range of vehicle types, ages, mileage levels, and condition scores. These dimensions allow the authors to test whether financial distress matters more for vehicles with longer expected service lives.

Table 1

Most common car characteristics

Brands				Models			
Rank	Car Make	# of Obs.	% of Obs.	Rank	Car Make	# of Obs.	% of Obs.
1	FORD	1,387,982	22.43	1	TAURUS	227,158	3.67
2	CHEVROLET	982,143	15.87	2	EXPLORER 4WD V6	123,892	2.00
3	NISSAN	351,425	5.68	3	IMPALA	118,306	1.91
4	TOYOTA	313,965	5.07	4	ALTIMA	112,353	1.82
5	PONTIAC	285,381	4.61	5	GRAND PRIX	108,419	1.75
6	JEEP	255,400	4.13	6	FOCUS	99,004	1.60
7	DODGE	216,632	3.50	7	F150 PICKUP 4WD V8	97,916	1.58
8	HONDA	199,190	3.22	8	MALIBU V6	66,717	1.08
9	B M W	170,190	2.75	9	F150 PICKUP 2WD V8	65,494	1.06
10	HYUNDAI	162,162	2.62	10	MUSTANG V6	64,639	1.04

Model Year				Category			
Rank	Year	# of Obs.	% of Obs.	Rank	Category	# of Obs.	% of Obs.
1	2006	1,278,944	20.67	1	SUV	1,839,362	29.73
2	2005	1,225,995	19.81	2	MIDSIZE CAR	1,486,014	24.02
3	2004	833,631	13.47	3	LUXURY CAR	759,971	12.28
4	2007	815,163	13.17	4	COMPACT CAR	706,807	11.43
5	2003	660,975	10.68	5	PICKUP	593,749	9.60
6	2002	385,805	6.23	6	VAN	464,510	7.51
7	2008	250,785	4.05	7	SPORTS CAR	184,297	2.98
8	2001	247,052	3.99	8	FULLSIZE CAR	124,620	2.01
9	2000	183,006	2.96	9	EXCLUDED	27,080	0.44
10	1999	124,512	2.01				

Car Condition			
Rank	Condition	# of Obs.	% of Obs.
1	3	3,622,619	58.54
2	4	1,310,341	21.17
3	2	918,274	14.84
4	1	174,978	2.83
5	5	141,234	2.28
6	0	21,313	0.34

The table presents the most common car characteristics by different variables in our sample. The % of Obs. refers to the % share of the category among observations for which information was available.

Table 1: Most common car characteristics

Table 2

Car condition levels

Grade	0	1	2	3	4	5
Paint & Body	Good for parts only	Sustained major collision damage, but may be drivable May be cost prohibitive to extensively recondition this vehicle by industry standards	Dents, scratches, and body panels require replacement Parts broken and missing	Conventional body and paint work needed Requires parts	Minor conventional body and paint work Small dents that have not broken the paint	No or minor defects
Interior	Missing or disconnected mechanical parts	Multiple prior repairs performed at substandard levels	Sustained cosmetic/light collision damage and repaired to industry standards	High quality conventional repairs of cosmetic/light collision damage		
	Operable, but near the end of its useful life Mechanical and body parts may be inoperable, disconnected, damaged, or missing	Repaired or unrepaired collision damage Operability of accessories is doubtful	Windshield may be damaged Signs of excess wear	Minor pitting of glass Signs of normal wear and usage	Minimal wear and minor missing or broken parts	Shows no signs of wear
Frame/Unibody		Burns, cuts, tears, and non-removable stains Repaired or unrepaired frame structure or frame damage	Requires repair or replacement of parts No repairs or alterations	No odors No repairs or alterations	No repairs or alterations	No repairs or alterations
Mechanical		Mechanical damage that prohibits operation properly Engine and/or transmission in poor condition	Mechanically sound Requires maintenance or minor repair of accessories	Sound and operable Fluids may require service or replacement	Mechanically sound Accessories are operable	Mechanically sound Accessories are operable
		Operability of accessories is questionable Average or better	Fluid levels low or require replacement Match by size and style	Identical Fixed or better condition	Identical Near new condition	Identical Near new condition
Tires		Worn or mismatched				

Table 2: Car condition levels

5.2 Figure 1 and Table 3: CDS distress and auction market structure

Read:

- Figure 1 shows CDS spreads for GM, Ford, Toyota, and Honda.
- GM and Ford CDS spreads rise sharply in 2008, indicating severe distress.
- Toyota and Honda have much lower CDS levels, although their spreads also rise in the crisis.
- Table 3 shows that 76.32% of auctions are open and 23.68% are closed.
- Most sales occur through physical lanes; fleet/lease and factory vehicles dominate the vehicle-source categories.

Economic message: The paper studies a liquid wholesale market with professional buyers, while using CDS spreads as timely measures of manufacturer default risk.

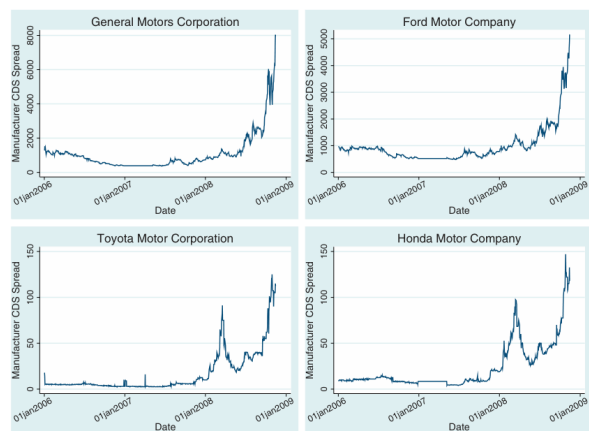


Figure 1: Auto manufacturer CDS spreads

Table 3
Auction characteristics

Auction Open/Closed			Vehicle Source			Transaction Type		
Closed?	# of Obs.	% of Obs.	Source	# of Obs.	% of Obs.	Type	# of Obs.	% of Obs.
N	4,723,193	76.32	Fleet/Lease	4,140,882	66.91	Lane	5,110,836	82.58
Y	1,465,566	23.68	Factory	1,741,028	28.13	Upstream	953,435	15.41
			Dealer	306,849	4.96	Online	124,488	2.01

Top 10 Auction Locations

Rank	Auction Location	# of Obs.	% of Obs.
1	Pennsylvania	474,288	7.66
2	Orlando	269,173	4.35
3	Riverside	225,562	3.64
4	Nashville	207,583	3.35
5	Dallas	203,371	3.29
6	Southern California	179,224	2.90
7	Chicago	165,734	2.68
8	New Jersey	157,914	2.55
9	Georgia	155,448	2.51
10	Milwaukee	154,819	2.50

Summary statistics of auction characteristics in our sample. The table contains information on the most common auction locations, whether auctions were closed to non-franchised dealers, the way purchases were transacted, and the source of the used vehicles.

Table 3: Auction characteristics

5.3 Tables 4–6: summary statistics, condition, and mileage

Read:

- The average transaction price is \$13,062 and the median is \$12,300.
- The average manufacturer CDS is 643 basis points, with a large standard deviation of 856 basis points.
- Average mileage is 44,270 miles.
- Price rises sharply with condition and falls sharply with mileage.
- The worst-condition cars have the highest average CDS exposure, while the best-condition cars have lower average CDS exposure.

Economic message: The data contain substantial price and distress variation, but the authors need detailed controls because quality and mileage are strongly related to prices.

Table 4
Summary statistics of select variables

Variable	Min.	Max.	Mean	Median	S.D.
Run #	1	3,960	185.15	121	227.64
# of Wheel-ins	0	80	0.30	0	0.92
Miles	1	999,991	44,270.38	31,743	36,875.77
Price	0	341,000	13,062.27	12,300	7,560.18
Manuf. CDS	2.5	8,039.70	643.13	520	856.14
# of Same Trim Cars That Day	0	443	8.98	2	19.42
# of Same Trim Cars So Far	0	443	3.39	0	9.44

Table 5
Car prices and manufacturer CDS by car condition

Condition	Avg. Price	Avg. Manuf. CDS
0	3,743.25	834.06
1	6,753.09	894.28
2	8,681.25	635.18
3	13,111.68	640.04
4	16,340.73	640.99
5	19,085.67	453.94

Table 6
Car prices and manufacturer CDS by mileage bands

Mileage Band	Avg. Mileage	Avg. Price	Avg. Manuf. CDS
1	4,818.98	19,823.08	614.32
2	10,244.34	17,936.70	649.72
3	13,244.71	17,111.88	625.50
4	15,880.88	16,573.01	609.78
5	18,459.50	16,143.92	603.33
6	20,856.87	15,730.07	616.69
7	23,214.74	15,149.07	627.93
8	25,765.59	14,497.46	638.18
9	28,215.15	13,917.72	701.23
10	30,416.64	13,901.36	691.40
11	33,417.61	14,193.16	644.36
12	37,167.25	14,303.58	591.99
13	41,836.43	13,882.19	570.29
14	47,083.13	13,417.75	553.98
15	53,753.09	11,985.07	578.45
16	62,320.51	10,172.16	616.94
17	73,232.08	8,152.84	678.44
18	87,000.30	6,265.74	710.17

Table 4: Summary statistics, car condition, and mileage bands

5.4 Figure 2: used car prices and CDS for Ford and GM

Read:

- The figure compares relative CDS movements and relative used car price residuals for Ford and GM versus Honda.
- In 2008, Ford and GM CDS spreads rise sharply and their relative used car price residuals fall.
- The visual evidence is clearest during the crisis period.
- The figure is motivating rather than identifying; the main regressions use higher-frequency within-week variation.

Economic message: The raw pattern is consistent with the paper's central hypothesis: rising manufacturer distress reduces the value of used cars from that manufacturer.

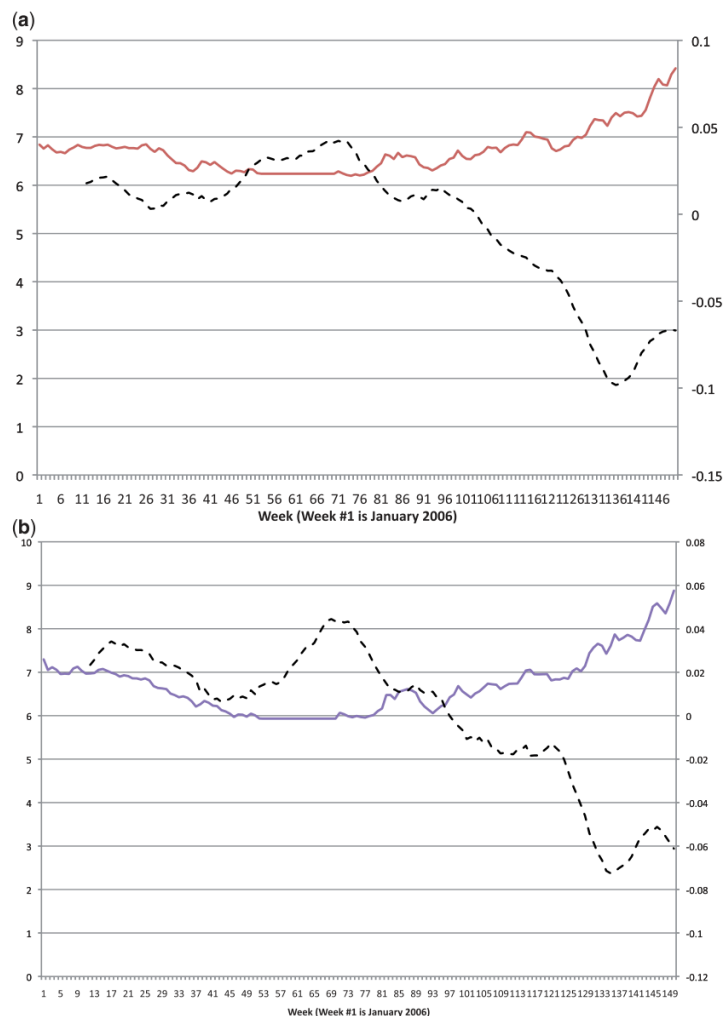


Figure 2
Average used car prices and CDS for Ford and GM
 The panels compare the relative average used car prices and CDS series for Ford (top) and GM (bottom). Each series shows the difference between the appropriate Ford (GM) series and the corresponding series for Honda. The price series are constructed by taking the residual from a regression of cars' auction prices on detailed car type fixed effects, sets of dummies for mileage quantiles, auction location fixed effects, and week-of-year

Figure 2: Average used car prices and CDS for Ford and GM

5.5 Table 7: baseline CDS effect on used car prices

Read:

- Column (1) estimates a CDS coefficient of -0.0679, significant at the 1% level.
- This means a 1,000-basis-point CDS increase lowers used car value by about \$68.
- Relative to the average price of \$13,062, this is about a 0.5% price drop.
- Column (3) confirms the result using normalized price.
- Mileage interactions show that the effect is concentrated in lower-mileage cars.

Economic message: The baseline estimate provides direct evidence that manufacturer distress is capitalized into durable-good prices.

Table 7
Effect of auto manufacturers' CDS spread on used car prices: Baseline specification

Dependent Var:	(1) Price	(2) Price	(3) Normalized price	(4) Normalized price
Manuf. CDS	-0.0679*** (0.0214)	-0.00514 (0.0340)	-5.15e-06*** (1.52e-06)	-3.58e-06 (2.69e-06)
Manuf. CDS × Band 2		-0.0419* (0.0233)		-5.60e-07 (1.18e-06)
Manuf. CDS × Band 3		-0.0608** (0.0250)		-1.11e-06 (1.19e-06)
Manuf. CDS × Band 4		-0.0854*** (0.0296)		-2.50e-06* (1.42e-06)
Manuf. CDS × Band 5		-0.105*** (0.0322)		-3.24e-06** (1.45e-06)
Manuf. CDS × Band 6		-0.119*** (0.0350)		-3.70e-06** (1.66e-06)
Manuf. CDS × Band 7		-0.128*** (0.0338)		-4.27e-06*** (1.47e-06)
Manuf. CDS × Band 8		-0.143*** (0.0382)		-5.30e-06*** (1.79e-06)
Manuf. CDS × Band 9		-0.149*** (0.0360)		-5.71e-06*** (1.78e-06)
Manuf. CDS × Band 10		-0.147*** (0.0381)		-5.94e-06*** (1.71e-06)
Manuf. CDS × Band 11		-0.120*** (0.0333)		-4.89e-06*** (1.66e-06)
Manuf. CDS × Band 12		-0.132*** (0.0296)		-6.45e-06*** (1.84e-06)
Manuf. CDS × Band 13		-0.108*** (0.0324)		-6.73e-06*** (2.57e-06)
Manuf. CDS × Band 14		-0.0811** (0.0350)		-7.11e-06** (3.02e-06)
Manuf. CDS × Band 15		-0.00877 (0.0427)		-5.63e-06 (4.43e-06)
Manuf. CDS × Band 16		0.0429 (0.0420)		-3.39e-06 (6.00e-06)
Manuf. CDS × Band 17		0.107** (0.0450)		3.41e-06 (6.92e-06)
Manuf. CDS × Band 18		0.151*** (0.0466)		1.33e-05* (7.93e-06)
Manuf. CDS × Band 19		0.158*** (0.0482)		2.09e-05** (9.94e-06)
Manuf. CDS × Band 20		0.173*** (0.0469)		3.43e-05*** (1.28e-05)
Band 2	-219.4*** (15.68)	-193.7*** (21.75)	-0.00632*** (0.000871)	-0.00602*** (0.00120)
Band 3	-418.3*** (18.33)	-380.2*** (23.39)	-0.0182*** (0.000975)	-0.0175*** (0.00129)
Band 4	-581.1*** (21.55)	-527.7*** (26.79)	-0.0284*** (0.00112)	-0.0268*** (0.00151)
Band 5	-741.9*** (22.95)	-676.2*** (28.08)	-0.0390*** (0.00117)	-0.0369*** (0.00157)
Band 6	-998.4*** (26.06)	-924.5*** (31.73)	-0.0565*** (0.00138)	-0.0542*** (0.00187)
Band 7	-1,116*** (27.17)	-1,037*** (32.69)	-0.0648*** (0.00143)	-0.0622*** (0.00190)
Band 8	-1,268*** (29.42)	-1,178*** (35.67)	-0.0758*** (0.00162)	-0.0725*** (0.00214)
Band 9	-1,412*** (33.40)	-1,318*** (39.10)	-0.0863*** (0.00205)	-0.0827*** (0.00257)
Band 10	-1,507*** (31.19)	-1,407*** (37.97)	-0.1000*** (0.00195)	-0.0961*** (0.00242)

(continued)

Table 7
Continued

Dependent Var:	(1) Price	(2) Price	(3) Normalized price	(4) Normalized price
Band 11	-1,767*** (29.85)	-1,692*** (37.40)	-0.111*** (0.00186)	-0.108*** (0.00233)
Band 12	-2,018*** (32.46)	-1,938*** (40.44)	-0.128*** (0.00206)	-0.124*** (0.00261)
Band 13	-2,454*** (35.62)	-2,387*** (45.40)	-0.157*** (0.00236)	-0.153*** (0.00300)
Band 14	-2,802*** (39.96)	-2,748*** (51.86)	-0.181*** (0.00277)	-0.177*** (0.00347)
Band 15	-3,491*** (44.64)	-3,476*** (58.85)	-0.231*** (0.00343)	-0.227*** (0.00421)
Band 16	-4,050*** (44.36)	-4,066*** (58.16)	-0.282*** (0.00454)	-0.280*** (0.00543)
Band 17	-4,657*** (44.50)	-4,721*** (57.39)	-0.353*** (0.00642)	-0.355*** (0.00733)
Band 18	-5,262*** (44.48)	-5,363*** (57.16)	-0.448*** (0.00926)	-0.457*** (0.0100)
Band 19	-5,818*** (44.35)	-5,927*** (57.70)	-0.566*** (0.0130)	-0.582*** (0.0136)
Band 20	-6,431*** (45.82)	-6,551*** (58.71)	-0.735*** (0.0173)	-0.761*** (0.0174)
Auction Closed	269.2*** (24.17)	275.5*** (24.17)	0.0245*** (0.00190)	0.0246*** (0.00189)
Condition 1	3,065*** (96.01)	3,065*** (95.98)	0.360*** (0.0153)	0.360*** (0.0154)
Condition 2	4,000*** (132.4)	4,004*** (131.7)	0.461*** (0.0213)	0.461*** (0.0212)
Condition 3	4,847*** (137.1)	4,851*** (136.4)	0.547*** (0.0229)	0.547*** (0.0229)
Condition 4	5,091*** (134.7)	5,095*** (134.1)	0.562*** (0.0228)	0.562*** (0.0227)
Condition 5	5,333*** (136.2)	5,335*** (135.7)	0.570*** (0.0226)	0.571*** (0.0226)
# of Same Trim Cars	1,964*** (0.367)	1,894*** (0.363)	6.17e-05** (2.47e-05)	5.92e-05** (2.44e-05)
That Day	-5,601*** (0.635)	-5,552*** (0.631)	-0.000268*** (3.89e-05)	-0.000266*** (3.89e-05)
# of Same Trim Cars So Far	-0.273*** (0.0164)	-0.272*** (0.0164)	-3.48e-05*** (1.58e-06)	-3.48e-05*** (1.58e-06)
Run #	-120.3*** (4.090)	-119.7*** (4.129)	-0.00930*** (0.000401)	-0.00927*** (0.000407)
# of Wheel-ins	10,768*** (142.5)	10,720*** (135.3)	0.668*** (0.0200)	0.667*** (0.0203)
Constant	6,188,759 0.986	6,188,759 0.986	6,188,725 0.899	6,188,725 0.899

The dependent variable is the (raw or normalized) transacted price of the used cars in the sample. *Manuf. CDS* refers to the credit-default swap (CDS) spread (in basis points) of the manufacturers of the used cars. *Manuf. CDS × Band 2-Band 20* denotes the interactions of *Manuf. CDS* with a set of dummy variables indicating to which of the 20 mileage bands a car belongs. *Auction Closed* is a dummy for whether the auction was closed to non-franchised dealers. *# of Same Trim Cars That Day* is the number of cars of the same model, trim and model year being sold on the same day in the particular auction location. *# of Same Trim Cars So Far* is the number of cars of the same model, trim, and model year being sold on the same day in the particular auction location before the auction. *# of Wheel-ins* is the number of times the car was wheeled through the auction lane. *Run #* is the location of the car in a particular run. Columns (1) and (3) also include car condition controls. All regressions include car type, region, week fixed effects. Reported standard errors are clustered on car manufacturers × month.

5.6 Table 8 and Figure 3: warranty and mileage channels

Read:

- Out-of-warranty cars have an insignificant CDS effect.
- In-warranty cars have a significantly more negative CDS effect; the implied full effect is about -0.062 dollars per CDS basis point.
- A car with full warranty remaining loses about \$139 per 1,000-basis-point CDS increase.
- Figure 3 shows that the effect is most negative for low- and medium-mileage cars and becomes much weaker for high-mileage cars.
- The negative effect is concentrated below roughly 50,000 miles, matching common warranty mileage cutoffs.

Economic message: The warranty and mileage evidence links the price drop to expected loss of future manufacturer-provided services, not merely to brand-level sentiment.

Table 8
Effect of auto manufacturers' CDS spread on used car prices: The warranty channel

Dependent var.:	(1) Price	(2) Price	(3) Price
Manuf. CDS	-0.00580 (0.0246)	-0.0101 (0.0322)	0.463*** (0.0522)
Car in warranty?	1,890*** (32.67)		
Car in warranty? $\times$ Manuf. CDS	-0.0565** (0.0233)		
Fraction of remaining warranty		4,145*** (89.42)	
Fraction of remaining warranty $\times$ Manuf. CDS		-0.129* (0.0716)	
Condition 1 $\times$ Manuf. CDS			-0.342*** (0.0358)
Condition 2 $\times$ Manuf. CDS			-0.415*** (0.0437)
Condition 3 $\times$ Manuf. CDS			-0.518*** (0.0510)
Condition 4 $\times$ Manuf. CDS			-0.527*** (0.0542)
Condition 5 $\times$ Manuf. CDS			-0.540*** (0.0690)
Condition 1			2956*** (81.05)
Condition 2			3972*** (100.7)
Condition 3			5019 (102.6)
Condition 4			5416*** (101.1)
Condition 5			5806*** (104.7)
Constant	7,414*** (124.9)	7,283*** (132.5)	8,152*** (99.80)
Observations	6,188,759	6,188,759	6,188,759
R-squared	0.982	0.982	0.979

The dependent variable is the transacted price of the used cars in the sample. *Manuf. CDS* refers to the credit-default swap (CDS) spread (in basis points) of the manufacturers of the used cars. *Car in warranty?* is an indicator variable denoting if a car is still under original factory warranty. This is also interacted with the manufacturer CDS. *Fraction of remaining warranty* is calculated as the minimum fraction between the mileage and age limits; cars out of warranty receive a value of zero. We also use the car condition indicators (0–6) defined in Table 2. Other controls not reported in the table include dummies for the auction format the car was sold under, the number of times the car was wheeled through the auction lane, and the number of cars of the same model, trim, and model year being sold on the same day in the particular auction location. All regressions also include car type-region-week fixed effects. Reported standard errors are clustered on car manufacturers $\times$ month, and are

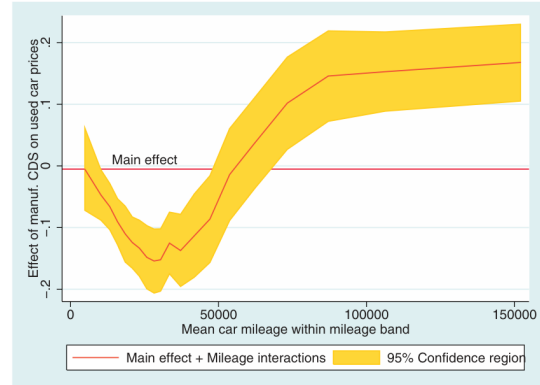


Figure 3
Plot of mileage interaction coefficients from Column (3) of Table 7

Table 5: Warranty channel

Figure 3: Mileage interaction coefficients

5.7 Tables 9–10: placebo, dealer distress, and new car prices

Read:

- Table 9 uses an unrelated manufacturer's CDS as a placebo and finds no significant effect.
- Table 9 also checks whether GM and Ford affiliated dealers change purchases when the manufacturer is distressed.
- Table 10 controls for contemporaneous new car prices.

- The warranty-channel estimates remain large and statistically significant after controlling for new car prices.

Economic message: The robustness tests reduce concern that the main effect is caused by broad auction supply shocks, dealer balance sheets, or new car fire sales.

Table 9
Robustness check: Placebo and dealer financial distress

	(1)	(2)	(3)
Dependent var.:	Price	GM dealer buys	Ford dealer buys
"Placebo" Manuf. CDS	0.047 (0.0758)		
Manuf. CDS		1.55e-05*** (4.89e-06)	-1.69e-05* (1.03e-05)
Constant	15447*** (33.10)	0.299*** (0.00551)	0.397*** (0.00982)
Observations	6,177,673	1,744,349	1,782,919
R-squared	0.984	0.523	0.513

The dependent variable in Column (1) is the transacted price of the used cars in the sample. "Placebo" Manuf. CDS refers not to the actual manufacturer CDS, but to the CDS of an unrelated manufacturer. In Columns (2) and (3), the dependent variable is an indicator for whether the buyer is a GM or a Ford dealer, respectively. These regressions use the CDS of the car's manufacturer. Other controls not reported in the table are the same as in Table 7 and include dummies for the auction format the car was sold under, the number of times the car was wheeled through the auction lane, and the number of cars of the same model, trim, and model year being sold on the same day in the particular auction location. All regressions also include car type-region-week fixed effects. Reported standard errors are clustered on car manufacturers×month, and are reported in parentheses. (***) denotes significance at the 1% level, ** at the 5% level, and * at the 10% level.)

Table 6: Placebo and dealer financial distress

Table 10
Robustness check: Controlling for new car prices

	(1)	(2)	(3)
Dependent var.:	Price	Price	Price
Manuf. CDS	-0.0775 (0.0599)	0.0843 (0.0647)	0.0677 (0.0572)
Car in warranty?		2038*** (50.28)	
Car in warranty? × Manuf. CDS		-0.135*** (0.0326)	
Fraction of remaining warranty			4721*** (134.6)
Fraction of remaining warranty × Manuf. CDS			-0.333** (0.152)
New car price	-0.0019 (0.0032)	-0.0023 (0.0033)	-0.0018 (0.0033)
Constant	17,102*** (88.47)	13,337*** (83.34)	13,333*** (78.52)
Observations	1,007,990	1,007,990	1,007,990
R-squared	0.980	0.975	0.977

Columns (2) and (3) of this table replicate Columns (1) and (2) of Table 8, but controlling for the new car price on the day of the auction (note that sample size is much smaller as we could not find new car transaction data for many auction days). Column (1) estimates the main effect of manufacturer CDS. As in Table 8, the dependent variable is the transacted price of the used cars in the sample. Manuf. CDS refers to the credit-default swap (CDS) spread (in basis points) of the manufacturers of the used cars. Car in warranty? is an indicator variable denoting if a car is still under original factory warranty. This is also interacted with the manufacturer CDS. Fraction of remaining warranty is calculated as the minimum fraction between the mileage and age limits; cars out of warranty receive a value of zero. Other controls not reported in the table include dummies for the auction format the car was sold under, the number of times the car was wheeled through the auction lane, and the number

Table 7: Controlling for new car prices

5.8 Table 11: implied costs of financial distress and capital structure

Read:

- Panel A maps credit ratings into CDS spreads and expected value loss.
- Estimated value loss rises from 0.11% for AAA firms to 7.17% for B-rated firms.
- Panel B shows marginal costs of debt rising sharply with lower credit ratings: 2% for AAA, 20% for BBB, and 39% for B.
- Panel C compares marginal tax benefits with marginal distress costs for GM and Ford.
- Over 1998–2007, GM's average marginal benefit minus marginal cost is approximately -1%, while Ford's is approximately -5%.

Economic message: The estimated indirect demand-side cost is large enough to matter for capital structure. For lower-rated auto manufacturers, distress costs can offset or exceed tax benefits of debt.

Table 11
Cost of financial distress and implications for capital structure

Panel A: Ex ante cost of financial distress

Credit rating	Risk-neutral probability of default (%)	CDS (bp)	Value loss (%)	Value loss (Almeida and Philippon 2007) (%)
AAA	0.54	32	0.11	0.32
AA	1.65	97	0.34	1.84
A	7.07	415	1.45	3.83
BBB	11.39	669	2.34	4.53
BB	21.07	1,237	4.33	6.81
B	34.90	2,049	7.17	9.54

Panel B: Marginal cost

Credit rating	dCDS / d ratings	Marginal cost (%)
AAA	48	2
AA	192	9
A	286	14
BBB	411	20
BB	690	33
B	812	39

Panel C: Marginal tax benefits vs. marginal cost

General Motors				
Year	Credit Rating	Marginal Benefit (effective tax rate) (%)	Marginal Cost (%)	Marginal Benefit – Marginal Cost (%)
1985	AA+	46	7	39
1986	AA	44	9	34
1987	AA–	37	11	27
1988	AA–	34	11	23
1989	AA–	34	11	23
1990	AA–	16	11	5
1991	A	0	14	–14
1992	A–	0	12	–12
1993	BBB+	28	18	10
1994	BBB+	33	18	15
1995	A–	35	12	23
1996	A–	35	12	23
1997	A–	35	12	23
1998	A	33	14	19
1999	A	35	14	21
2000	A	34	14	20
2001	BBB+	20	18	3
2002	BBB	18	20	–2
2003	BBB	30	20	10
2004	BBB–	33	24	9
2005	B	19	39	–20
2006	B	5	39	–34
2007	B	4	39	–36
Avg. 1998–2007	N/A	23	24	–1
Ford				
1985	A	46	14	32
1986	AA–	46	11	35
1987	AA–	39	11	28
1988	AA	34	14	20
1989	AA	34	14	20

(continued)

Table 11
Continued

Panel C: Marginal tax benefits vs. marginal cost

Ford				
Year	Credit Rating	Marginal Benefit (effective tax rate) (%)	Marginal Cost (%)	Marginal Benefit – Marginal Cost (%)
1990	AA–	32	11	21
1991	A	17	14	3
1992	A	0	14	–14
1993	A	35	14	21
1994	A	35	14	21
1995	A+	35	7	28
1996	A+	35	7	28
1997	A	35	14	21
1998	A	35	14	21
1999	A+	35	7	28
2000	A	25	14	11
2001	BBB+	7	18	–11
2002	BBB	22	20	2
2003	BBB–	5	24	–19
2004	BBB–	2	24	–23
2005	BBB+	10	29	–19
2006	B	2	39	–37
2007	B	35	39	–4
2008	CCC+	1	39	–38
2009	B–	31	39	–9
2010	B+	33	37	–4
Avg. 1998–2007	N/A	18	23	–5

Panel A computes the ex ante cost of financial distress for car manufacturers with different credit ratings. *Risk-neutral probability of default* is the five-year risk-neutral probability of default from Almeida and Philippon (2007). Table 3. *CDS* is the expected CDS in basis points. *Value loss* is the expected share of value lost, in percent. The last column presents estimates of value loss from Almeida and Philippon (2007). Table 4, benchmark model. Panel B computes the marginal cost. *dCDS/d ratings* is computed as the average change in CDS from adjacent ratings. *Marginal cost* is the marginal cost of financial distress in percent of firm value. Panel C compares the marginal tax benefit of debt to the marginal cost for General Motors and Ford. Credit ratings are from Compustat. Effective tax rates are the effective tax rates after interest deductions provided by John Graham (see Graham 2000). For ratings below B, we assign the marginal cost at B. We interpolate for ratings that are not in the table above linearly.

6 Short summary

- The paper provides direct evidence of indirect costs of financial distress in a durable-goods market.
- The core idea is that manufacturer distress threatens warranties, parts, maintenance, and service networks.
- The setting is U.S. wholesale used car auctions from 2006 to 2008.
- The sample contains more than 6 million successful transactions.
- Manufacturer distress is measured using five-year CDS spreads.
- The main design uses car-type-region-week fixed effects and high-frequency CDS movements.
- A 1,000-basis-point CDS increase reduces the average used car price by about \$68.
- Cars under warranty and cars with more warranty remaining have much larger price drops.
- A full factory warranty loses about \$139 in value per 1,000-basis-point CDS increase.
- The mileage evidence shows that the CDS effect is concentrated among low-mileage cars with longer expected service lives.
- Placebo CDS spreads do not generate the same result.
- Dealer distress and new car prices do not explain the result.
- The estimates imply large indirect financial distress costs for car manufacturers.
- For lower-rated manufacturers, these costs can be several percent of firm value.
- The simple trade-off exercise suggests that debt tax benefits can be outweighed by indirect distress costs.
- The paper’s broader lesson is that product-market demand can be directly harmed by the mere possibility of financial distress.

7 Conclusion

7.1 Core conclusion

Hortaçsu, Matvos, Syverson, and Venkataraman show that financial distress reduces durable-goods values before bankruptcy occurs. In the auto market, buyers discount used cars from distressed manufacturers because they fear that warranties, spare parts, maintenance, and dealer support may be disrupted. This creates a measurable indirect cost of financial distress.

7.2 Economic intuition

A durable good is not just the physical product. It is a bundle of future service flows. A car is more valuable when the owner expects the manufacturer and dealer network to honor warranties, supply parts, and provide service. When a manufacturer’s default probability rises, the future service bundle becomes riskier. Rational buyers capitalize this risk into today’s auction price.

7.3 Takeaway

The paper gives a clear transaction-level demonstrations that financial distress can damage product-market demand. The result matters for capital structure because taking on more debt does not merely increase expected bankruptcy costs; it can also lower consumers’ willingness to pay for the firm’s products while the firm is still operating.